AI61002 : Deep Learning Foundations and Applications
Spring 2025

Assignment 4: Transformer Models
(20 Marks)

Due Date: 18/04/2024

Instructions:

Complete the tasks listed below, ensuring you follow the specified instructions for each. Use PyTorch for implementation and provide step-by-step outputs in a Jupyter Notebook (.ipynb file). All relevant code snippets, visualizations, and explanations must be included within the notebook. Submit the required files as per the submission guidelines provided.

Task 1 - 3 Marks

Load the SST2 dataset from torchtext.datasets for the fine grained sentiment classification task (train and test splits). Show some samples from the dataset and corresponding class labels.
Preprocess the dataset to perform the following:

- Removing stopwords
- Removing punctuations
- Tokenization (using BERT Tokenizer)
- Converting text tokens to numerical data (using torchtext.vocab)
- One hot encoding of labels

Task 2 - 2 Marks

Split the train split into training and validation partitions and show the label distribution on the training set.

Task 3 - 5 Marks

1. Design a transformer encoder layer as done in class tutorial (i.e. using nn.MultiheadAttention, nn.Linear and nn.Layernorm with residual connection).
2. Design a transformer model using linear embedding layer (nn.Embedding), sinusoidal positional encoding, and the *N* number of transformer encoder layer designed above. Add a linear layer after the last encoder layer to perform classification.
3. Add dropouts after the linear embedding, after the multi-headed self attention and after the positionwise feedforward neural network for regularization.
4. Train the transformer model and show the variation of training and validation losses.

5. Evaluate the transformer model on the test set and report the performance (accuracy, precision, recall).

Task 4 - 4 Marks

1. Now replace the custom transformer encoder module in your transformer model with nn.TrasformerEncoderLayer() and train the model again. Show the variation of training and validation losses (note: use the same number of encoder layers and hidden state dimension of each layer as in the previous case).
2. Report the performance of the trained model using nn.TransformerEncoderLayer() on the test (accuracy, precision, recall).

Task 5 - 5 Marks

1. Perform transfer learning by using the BERT encoder as a pre-trained encoder and train a linear layer on the encoder representations (as done in the class tutorial). Show the training and validation losses for the transfer learning model.
2. Report the test set performance obtained with the above transfer learning.

Task 6 - 1 Marks

Show a summary of all models and summarize insights.

Hyperparameters

Common across all models:

- Learning rate: 1e-3
- Optimizer: Adam
- Loss function: CrossEntropyLoss
- Batch size: 32
- Number of epochs: 5

Transformer-based models (custom and built-in):

- Hidden dimension (d_model): 16
- Number of attention heads (nhead): 4
- Number of layers in positionwise feedforward neural network: 1
- Number of encoder layers (num_layers): 1
- Dropout rate: 0.5
- Maximum sequence length for positional encoding (max_len): 5000

BERT-based model:

- Pretrained model: bert-base-uncased
- Freeze BERT encoder: True
- Maximum sequence length for tokenization: 64

Submission Guidelines

- The content that you submit must be your individual work.
- Submit both .py and .ipynb file formats. Both submissions are mandatory for credit.
- Name the files as "<roll_number>_assignment_4.py" and "<roll_number>_assignment_4.ipynb". Submit them in a single zip file named "<roll_number>_assignment_4.zip". For example, if your roll number is 23AI91R01, the submission file should be named 23AI91R01_assignment_4.zip. (Note: Do not submit .rar files.)
- Ensure all code is well-commented, and explanations are included using text cells in the Jupyter Notebook.
- Submit your assignment through Moodle before the deadline. Late submissions or email submissions will not be accepted.

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